Where is the tractogram redundancy?

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Introduction: Tractogram redundancy is well documented[5,6], but its distribution at the anatomical bundle level is not. We characterize per-bundle redundancy associated with fiber bundles extracted with TractSeg (TS)[8] for long-range bundles and SUP-WMA (SWM)[9] for superficial ones, and the per-cluster composition of foundational streamlines not modeled by either method.

Methods: We analyzed five preprocessed HCP subjects[3] with 10 M-streamline probabilistic IFOD2 tractograms each. Per-streamline consensus acceptance weights, $a \in [0, 1]$, were generated from randomized SIFT (rSIFT)[4] and randomized COMMIT (rCOMMIT)[7] with anatomical labels from TS (71 long-range bundles) and SWM (198 ORG-800FC clusters) to identify which pathways were associated with foundational or redundant streamlines. We defined streamlines as *foundational* if both weights $a \geq t$ and *redundant* if $0 < a < t$ in both filters; requiring both filters to agree makes the labels robust to single-filter artifacts. We used $t = 0.9$ in the experiments. For the streamlines not covered by TS and SWM, we used QuickBundles[2] with the Minimum average Direct-Flip distance (MDF) and a 20 mm threshold to test whether they formed coherent clusters. To understand where the unclassified streamlines accumulated, we stratified every streamline by atlas classification (TS±, SWM±) and acceptance pool (foundational $a \geq t$ vs. partial-acceptance redundant $0 < a < t$), and reported per-cell counts, mean length, and the top contributing TS bundles. We tested whether the foundational rate was uniform across the 71 TS bundles and the 198 SWM clusters using one-way ANOVA and Kruskal-Wallis tests on per-subject per-anatomy rates.

Results: Inter-bundle variation (Table 1, upper). The per-bundle foundational fraction varied strongly across the TS bundles, where small limbic tracts (fornix, uncinate, anterior commissure) carried the largest foundational fraction, while large projection and callosal tracts (corticospinal, frontopontine, mid-body corpus callosum) were nearly fully redundant but still carried tails of partial weights. rSIFT decayed faster than rCOMMIT, and the two diverged near the ventricles, where COMMIT was more affected by CSF partial volume effects[1].

SWM recovers part of the U-fiber streamlines (Table 1, lower). The majority of foundational streamlines were not covered by TS. QuickBundles showed they formed coherent short-range populations in occipital, frontal, and cerebellar cortex (median length ∼48 mm). Adding SWM reduced the uncovered foundational fraction by ∼40%. The ANOVA showed the foundational rate was not uniform over either atlas, varying significantly across both the 71 TS bundles ($F = 5.6$, $p \approx 10^{-25}$) and the 198 SWM clusters ($F = 7.8$, $p \approx 10^{-99}$).

Where redundancy is (Fig. 1). The partial-acceptance pool ($0 < a < 0.9$ in both filters; 32–38% of each tractogram, 79–83% TS-unclassified) traced the same anatomical paths as foundational streamlines but extended into the cortical periphery: redundant streamlines were ∼20 mm longer (77 vs 54 mm) and sat at a cross-subject median of 4.5–5.1 mm MDF from the nearest foundational streamline.

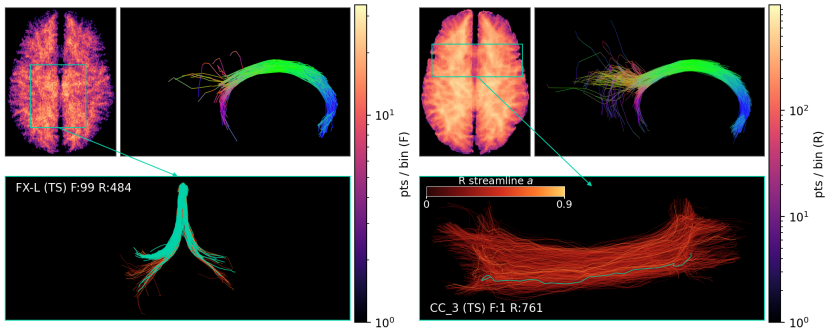


Figure 1: Where redundancy lives (subj 837964). *Top:* whole-brain axial density of F (left) and R (right) streamlines. *Bottom:* the corresponding bundle, F=green ($a \geq 0.9$ in both filters), R colored by mean acceptance weight $\bar{a} \in (0, 0.9)$ (dim→bright).

Table 1: Redundancy summary (5 subjects, mean±95% CI). SWM regions are identified by streamline midpoint.

Most foundational (top 3, foundational fraction)		
TS	FX-L, UF-R, CA	3.3–4.9%
SWM	c733 cb, c71 oc, c90 oc	11.3–16.4%
Most redundant (top 3, redundant fraction)		
TS	FPT-R, CC_4, FPT-L	>99.9%
SWM	c107 oc, c6 mix, c390 fr	98.9–99.0%

Foundational (F)/redundant (R) ± 95% CI	*F*	*R*
% of tract.	0.76 ± 0.25	36.1 ± 3.4
Length (mm)	54.0 ± 1.0	76.9 ± 2.8
TS-unclass. (%)	92.0 ± 2.3	80.8 ± 2.4
MDF to F (mm)	—	4.79 ± 0.27
Top R bundles	—	MLF, ST_PRF, CC_5

Conclusion: Tractogram redundancy varies with white-matter anatomy in this HCP cohort: a few small limbic tracts carry most of the foundational streamlines, while large projection and callosal tracts, together with the unclassified streamlines surrounding them, carry the bulk of the redundancy. The cross-subject spatial map and the ∼5 mm MDF between partial-acceptance and foundational streamlines characterize this redundancy as within-bundle over-production rather than mistracking: probabilistic tracking generates many alternate streamlines along the same anatomical paths, and signal-based filters retain only as many as the diffusion signal supports. Adding SWM roughly halves the unclassified foundational fraction (from ∼92% to ∼55%) but does not reduce this within-bundle redundancy, with a short-range orphan residual remaining unrecognized by either atlas.

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